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Github
Blog

Hyeongseob Jo

BIO

I am a Machine Learning Engineer at ML2. My graduate research focused on computer vision for automated manufacturing defect classification. I am currently working on multimodal learning at ML2.

FIELD OF INTEREST

Computer Vision, Defect Classification, Representation Learning, Multimodal Learning

EDUCATION

Dongguk University *Feb. 2026*
M.S. in Computer Science and Artificial Intelligence

Dongguk University *Feb. 2024*
Bachelor of Science in Engineering in Mechanical Engineering

Andong National University *Feb. 2020*
Bachelor of Arts in Ethics Education; Double Major in English Education

RESEARCH EXPERIENCE

Multi-Illumination Display Inspection *Mar. 2024 – Feb. 2025*
Smart Vision & Media (SVM) Lab

- Conducted as part of an Industry–Academic Cooperation Project with LG Display.
- Developed a defect classification algorithm for display panels using multi-illumination images.
- Achieved 98.5% classification accuracy and demonstrated applicability in an industrial setting.

POSITIONS

Machine Learning Engineer *Dec. 2025 – May. 2026*
ML2
Deploying Vision-language model on iOS devices using ONNX and Core ML.

Graduate Student Research Assistant *Mar. 2024 – Feb. 2026*
SVM Lab
Conducted graduate-level research in computer vision.

Contract Teacher (Part-Time) *Mar. 2020 – Dec. 2020*
Youngju Girls' High School
Taught Integrated Social Studies (Grade 1), Philosophy, Ethics (Grade 2)

PROJECT

Google Summer of Code Summer 2026
Project: Agentic AI for Predictive Maintenance using OpenVINO

- Selected as a contributor for Google Summer of Code (GSoC) 2026.
- Program duration: May 25 – Aug. 24, 2026.

Samsung Collegiate Programming Challenge Summer 2025
AI Competition: Multimodal AI for Everyday Photo Understanding

- Proposed CLIP · SigLIP-based ensemble model for multimodal image-text classification.
- Achieved 71% accuracy on test set (851 samples), ranked 41st out of 242 participants (Top 50 finalist).

LG Aimers Winter 2023
AI Competition: B2B Sales Opportunity Prediction

- Developed a binary classification model to predict B2B sales opportunities from MQL data.
- Benchmarked 10+ ML models (e.g., XGBoost, LightGBM) for B2B sales opportunity prediction.

Autonomous Robot Navigation Spring 2023
Dongguk University Course Project (MEC4092-01)

- Implemented ROS2-based autonomous navigation with real-time control and sensor-fusion.
- Evaluated navigation performance in a physical testbed environment with obstacle avoidance tasks.

PUBLICATIONS

International Conferences

1. Hyeongseob Jo, Seunggi Park, and Sung In Cho. A Survey of Unsupervised Learning-Based Out-of-Distribution Detection. In *Proceedings of the IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, Nov. 2024
2. Y. S. Oh, M. G. Kim, B. Kim, J. Y. Kim, Hyeongseob Jo, J. H. Park, G. Lee, and S. I. Cho. Dual-Branch Network via Multiple Illumination-Aware Representation Learning for Steel Surface Defect Classification. In *Proceedings of the British Machine Vision Conference (BMVC)*, Sheffield, UK, Nov. 2025
3. Yoonseo Park, Hyeongseob Jo, and Sung In Cho. A Survey of Training-free Diffusion-based Image Generation with Free-form Mask. In *Proceedings of the 2025 International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, Jul. 2025

Domestic Conference (Korean)

1. 이선호, 남혁, 조형섭, and 조성인. 차선 검출: 주요 접근법과 최신 연구 동향. In *대한전자공학회 하계학술대회 논문집*, pages 2969–2973, 2025

Magazine Article (Korean)

1. 조형섭, 오용석, and 조성인. 다중 조명 환경에서 제품 외관 검사 기법에 대한 이해. *Information Display (한국정보디스플레이학회)*, 26(6), Dec. 2025

Dissertations

1. 조형섭. 산업 공정의 다중 조명 환경에서 표면 결함 자동 검사를 위한 경량 딥러닝 기반 학습방법. Master's thesis, Dongguk University, Seoul, Korea, Feb 2026

PATENTS

Domestic Patent

[1] 조성인, 조형섭, 이미지 내 결함을 분류하는 시스템 및 분류 방법, 출원 *10-2025-0155364* (2025-10-24).

TEACHING ASSISTANT

AI course (machine learning, deep learning) *Jun. 10–13 and 23–25 2025*
LG Display

Introduction to Industry-Academia Project (CAI7177-01) *Spring 2025*
Computer Science and Engineering, Dongguk University

Signal Processing, Image Processing *Aug. 19–23 2024*
LX Semicon

Creative Engineering Design (CSC2004-03) *Spring 2024*
Computer Science and Engineering, Dongguk University

LANGUAGES & SKILLS

- Korean (native), English (conversational)
- PyTorch, MATLAB, ONNX, Core ML
- LaTeX, Microsoft Office, Linux (Ubuntu), Windows

Revised May 11, 2026